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DEEP LEARNING-BASED DEEPFAKE DETECTION

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ABSTRACT

The fast development of deep learning technologies has made it possible to create very realistic manipulated videos called deepfakes. These videos pose serious risks in areas such as misinformation, identity theft, and cybersecurity. This project offers a deep learning method to detect deepfake videos by looking at both spatial and temporal features. The system processes input videos by extracting frames, detecting facial regions, and producing sequences focused on faces. A Convolutional Neural Network (CNN), specifically ResNeXt-50, extracts spatial features from individual frames. Meanwhile, a Long Short-Term Memory (LSTM) network identifies temporal inconsistencies across frames. The model is trained on a balanced dataset of real and fake videos gathered from various sources. Experimental results indicate that the proposed approach has effective classification performance, with temporal modeling greatly enhancing accuracy. The system shows the promise of combining CNN and LSTM architectures for strong deepfake detection and can be expanded for real-time and multimodal analysis in future work.

1. INTRODUCTION

Manipulated content has grown more prevalent due to the quick expansion of digital media and social networking sites. The emergence of deepfakes—synthetic media produced using deep learning techniques—is among the most alarming trends. These films can effectively mimic a person's voice or face, making it challenging to distinguish between authentic and fraudulent content based just on human observation.

Generative Adversarial Networks (GANs) and encoder-decoder architectures are two common models used to create deepfakes. These technologies pose significant risks even if they can be useful for media creation and entertainment. Misinformation, financial fraud, identity theft, and reputational harm can result from the improper use of deepfakes. Identifying this type of modified content has emerged as a major cybersecurity concern.[7]

The increasing complexity of deep learning models frequently makes traditional techniques for identifying altered media ineffective. Automated detection systems that can spot minute irregularities in videos are therefore becoming more and more necessary.[1] By analysing patterns that are invisible to the human eye, artificial intelligence (AI), particularly deep learning, is essential to solving this problem.

The objective of this project is to develop an effective system that uses both temporal and spatial data to distinguish between real and deepfake films. The system looks for discrepancies between video frames by integrating Long Short-Term Memory (LSTM) networks for sequence analysis with Convolutional Neural Networks (CNNs) for feature extraction.

This project's primary goals are:

- To comprehend the creation of deepfakes
- To create a deep learning-based detecting system
- To examine features at the frame and sequence levels
- Using common measures to assess the model's performance

By providing a useful method for identifying altered video and reducing the dangers associated with deepfake technology, this initiative advances cybersecurity. Additionally, it supports the necessity of integrating CNN and LSTM in our research.

2. LITERATURE REVIEW

S.No	Author(s)	Method Used	Advantages	Limitations
1	Abhijit Jadhav, Abhishek Patange, Jay Patel, Hitendra Patil, Manjushri Mahajan[1]	CNN (ResNeXt) + LSTM for spatial and temporal analysis	Captures both frame-level and sequence-level features	Does not consider audio deepfakes
2	Andreas Rössler, Davide Cozzolino, Luisa Verdoliva, Christian Riess, Justus Thies, Matthias Nießner[2]	XceptionNet-based deepfake detection	High accuracy on benchmark datasets	Ignores temporal information
3	Yassine Afchar, Vincent Nozick, Junichi Yamagishi, Isao Echizen[3]	MesoNet (lightweight CNN)	Fast and computationally efficient	Lower accuracy on high-quality deepfakes
4	Yuezun Li, Siwei Lyu[4]	CNN-based artifact detection	Detects warping artifacts effectively	Fails on advanced deepfakes
5	Yuezun Li, Ming-Ching Chang, Siwei Lyu[5]	Eye blinking detection using LRCN	Uses biological signals	Modern deepfakes mimic blinking
6	Huy H. Nguyen, Junichi Yamagishi, Isao Echizen	Capsule Networks	Captures spatial relationships better	High computational complexity
7	Umur Aybars Ciftci, Ilke Demir, Lijun Yin	Biological signal analysis (PPG + CNN/SVM)	Works across video qualities	Complex and not real-time
8	David Güera, Edward J. Delp[6]	CNN + RNN for temporal analysis	Uses sequence modeling	Limited dataset

9	Brian Dolhansky, Joanna Bitton, Ben Pflaum, Jikuo Lu, Russ Howes, Menglin Wang, Cristian Canton Ferrer[9]	DFDC dataset-based detection	Large-scale real-world dataset	Requires high compute
10	Ekraam Sabir, Jia Wang, Anuj Tyagi, Avishek Majumdar, Zongwei Wang, Zhisheng Wang, Anil N. Bhanu[10]	CNN + RNN hybrid approach	Combines spatial + temporal features	Increased model complexity

According to the aforementioned research, deepfake detection techniques have been divided into three groups: biological signal-based techniques, temporal-based techniques, and spatial-based techniques.

CNN-based techniques are spatially based, while sequence models like RNN and LSTM are temporally based.

It is evident from the aforementioned techniques that the most of them have limits when it comes to identifying deepfake films. For instance, temporal inconsistencies are not considered in spatial-based approaches. Furthermore, a lot of data and a lot of processing power are needed for temporal-based approaches.

Despite the fact that numerous techniques have been put out to identify deepfake videos, the following restrictions still exist:

- Absence of models that take temporal and geographical characteristics into account
- Models' limited applicability to real-world data
- High model complexity in modern approaches
- Reduced emphasis on creating effective detection techniques

This study presents a hybrid deep learning strategy to detect deepfake films in order to address the shortcomings of the current methods. To improve detection, the suggested approach combines CNN and LSTM.

3. PROBLEM STATEMENT

Deepfakes are extremely convincing edited videos that have been created as a result of the quick development of deep learning technology, especially GANs. Deepfakes can be used for a number of manipulations, such as altering a person's voice, visage, and facial expressions, making it very challenging for an individual to discern between authentic and fraudulent material. By taking into account the potential misuse of this kind of media, the significance of identifying deepfake media may be comprehended. Deepfake media can be used to manipulate public opinion, disseminate false information, and harm people's reputations. In domains such as communication, law enforcement, and journalism, failing to discern between authentic and fraudulent material can have catastrophic outcomes. Thus, creating a trustworthy automated system to identify deepfakes is crucial.

Deepfake detection is a challenging task for a number of reasons. First of all, the most recent methods for producing deepfake media are very sophisticated, enabling the production of excellent media with few visual artefacts, which makes it very challenging for current media detection methods to function. Second, spatial features are the main emphasis of current media detection methods, which can be problematic because temporal elements are not taken into account. Lastly, it is very challenging for an individual to employ some of the most recent methods for media detection because they are very sophisticated and require biological signals.

The problem of detecting deepfakes can be defined as designing and implementing a highly efficient deep learning-based media detection system capable of distinguishing between real and fake media by analyzing spatial and temporal features.

4. PROPOSED METHODOLOGY

The suggested technique uses temporal and geographical characteristics to identify deepfake films. While temporal features will identify discrepancies between two successive video frames, spatial features will assist in identifying visual artefacts within the film.

Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks will both be used in the hybrid deep learning architecture of the suggested system. By using both spatial and temporal information for deepfake video identification, this will raise the system's accuracy level.

4.1 System Architecture

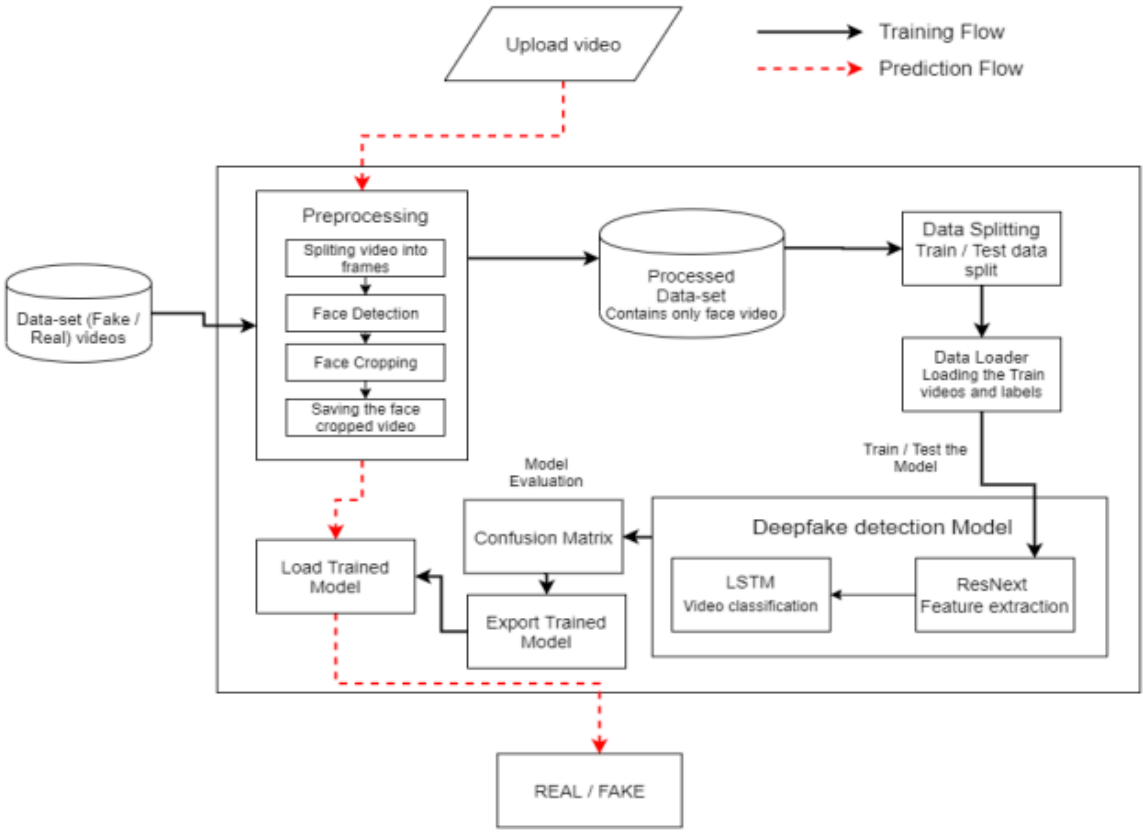

Figure 4.1.1 : System Architecture

4.2 AI/ML Models Used

The suggested system would employ a hybrid deep learning approach that combines Long Short-Term Memory (LSTM) networks with Convolutional Neural Networks (CNN). This would raise the suggested system's accuracy level because it will employ temporal and spatial information to detect deepfake videos.[1] [6]

A pre-trained ResNeXt-50 CNN model will be utilised by the suggested system to extract spatial features from the video. Inconsistencies generated during the deepfake video

generation process can be detected by the ResNeXt-50 model. Face deformation, texture, and lighting could be the cause of these discrepancies.

The LSTM model is then applied to the high-level characteristics that were retrieved from each frame. The system can comprehend the temporal relationship between each pair of subsequent frames with the aid of the LSTM model.

After that, the fully connected layer receives the data from the LSTM model. The acquired data is then subjected to the softmax algorithm. Based on the values acquired, the video is categorised as either real or deepfake.

4.3 Workflow of the Proposed System

Entering the video into the system is the first step in the suggested system's workflow. After that, the movie goes through a number of phases to extract the necessary data. The suggested method starts by splitting the video into multiple separate frames. After that, the faces found in each frame are trimmed by eliminating extraneous background information. This enables the system to focus solely on the necessary data.

A certain number of frames are extracted from the video for this purpose. After that, the retrieved frames are resized to 112 × 112 pixels. The ResNeXt-50 CNN model is then applied to each individual frame. The CNN model aids in the system's comprehension of each frame's high-level characteristics.

The LSTM model is then applied to the high-level characteristics that were retrieved from each frame. The system can comprehend the temporal relationship between each pair of subsequent frames with the aid of the LSTM model.

After that, the fully connected layer receives the data from the LSTM model. The acquired data is then subjected to the softmax algorithm. Based on the values acquired, the video is categorised as either real or deepfake.

4.4 Data Preprocessing

Preprocessing of the data is a very important step for the improvement of the reliability of the proposed model. The dataset for this project is collected from various sources, such as FaceForensics++, Deepfake Detection Challenge (DFDC), and Celeb-DF. To avoid bias, the dataset is well balanced, i.e., there are equal numbers of real and fake videos. Each of these videos is divided into individual frames, and then face detection techniques are applied to obtain only the face regions of the images. The images that do not contain faces are discarded. To avoid the limitations of the computational resources and for uniformity, a certain number of frames are taken from all the videos, i.e., a maximum of 150. The entire dataset, i.e., all the extracted images, are then resized to a size of 112 x 112 pixels and then used for training the network. The dataset is divided into the training and testing sets in the ratio of 70:30, ensuring that both sets contain equal proportions of real and fake images.

5. PERFORMANCE RESULTS

5.1 Test Bed Details

An experimental setting was used to implement the suggested system. Python was used as the system's programming language, and the implementation was carried out on a Linux environment. The PyTorch framework, which is used to create deep learning models, was used for the implementation.

To increase the computational efficiency of the suggested system, the tests were carried out using a laptop with 16 GB RAM and an NVIDIA RTX 3050 GPU. Although it would take a long time to run on a system without a GPU, the suggested solution can run on any system, GPU or not. Because the suggested system is platform-neutral, it can also be used in a Windows environment.

The proposed system was trained using batch processing, which is used to improve the performance of the proposed system using the Adam optimizer.

5.2 Dataset Details

A variety of publicly accessible deep fake datasets make up the dataset used to train and assess the model. These datasets include Celeb-DF, Deep Fake Detection Challenge (DFDC) [9], and Face Forensics++ [2]. These datasets' variety of real and false videos improves the model's capacity for generalisation.

The dataset has about 6000 videos in total. To avoid bias during training, an equal amount of real and phoney videos are used. 30% of the dataset is used for testing and 70% is used for training. Because of the diversity of the datasets, the model is exposed to various deepfake generating techniques and video attributes.

5.3 Evaluation Metrics

Standard classification criteria such as accuracy, precision, recall, and F1-score were used to evaluate the performance of the suggested model. By counting the number of accurate predictions, the accuracy metric evaluates the model's overall soundness. The number of predicted bogus videos that are indeed fake is known as precision, and it indicates the model's capacity to prevent false alarms. The number of real false videos that are accurately identified as such is known as recall, and it indicates how well the model can identify fraudulent videos. The model's performance in situations when both false alarms and false negatives are crucial is evaluated using the F1-score, which is defined as the harmonic average of precision and recall.

5.4 Results Table

The performance of the model was evaluated for different frame sequence lengths to analyze the impact of temporal information on detection accuracy. The results are summarized in the table below.

Frame Length	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
10 Frames	83.18	83.02	83.35	84.87
20 Frames	84.04	83.87	84.21	85.89
40 Frames	86.74	86.66	86.89	88.09
60 Frames	86.28	86.17	86.44	87.85
80 Frames	86.59	86.49	86.74	88.07
100 Frames	86.74	86.64	86.90	88.28

Table 5.4.1 : Result

5.5 Result Visualization

Visual representations are necessary to comprehend the model's performance. To comprehend genuine positives, false positives, true negatives, and false negatives, one can utilize a confusion matrix. This will provide a more thorough understanding of the classification's performance. A more thorough understanding of the model's learning and convergence will be provided by the accuracy vs. epochs graph.

To demonstrate how the system classifies the films as authentic or fraudulent, sample prediction results should be provided.

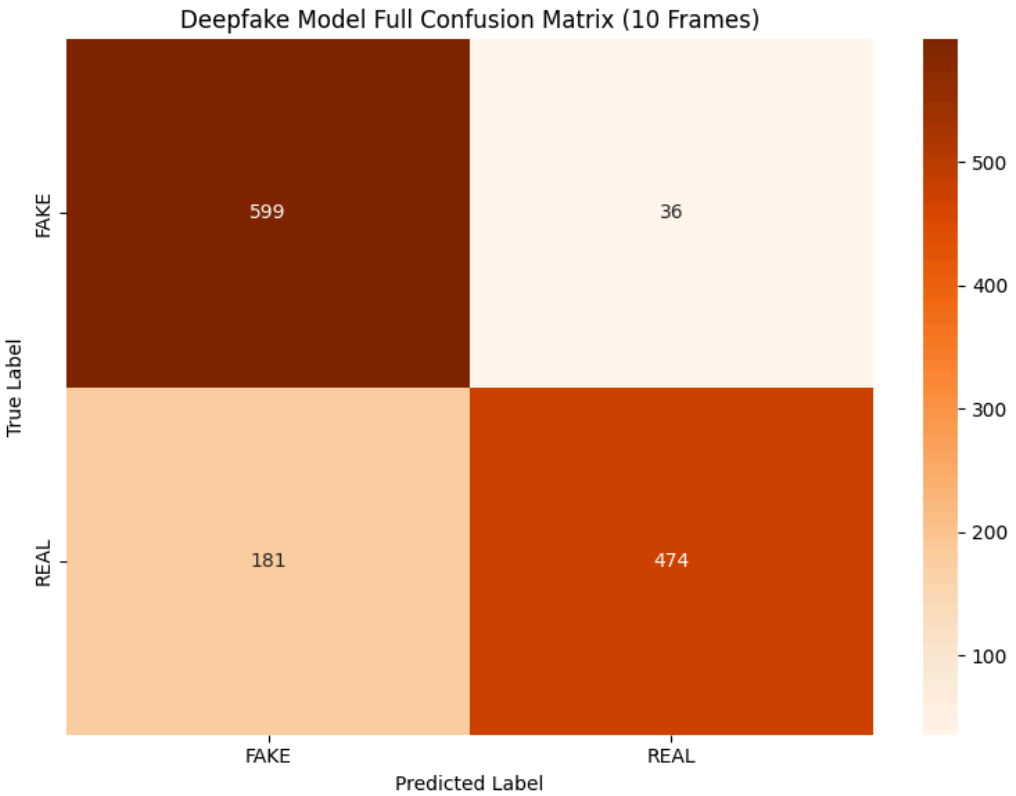

Figure 5.5.1 : Confusion Matrix (10 frames)

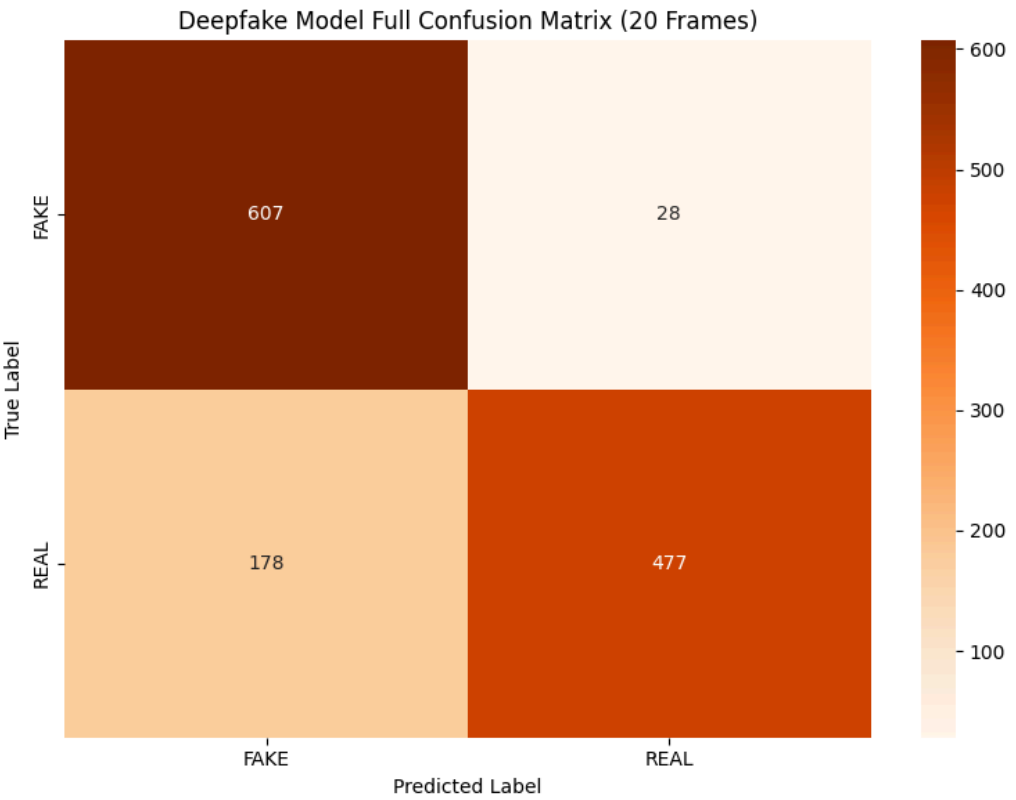

Figure 5.5.2 : Confusion Matrix (20 frames)

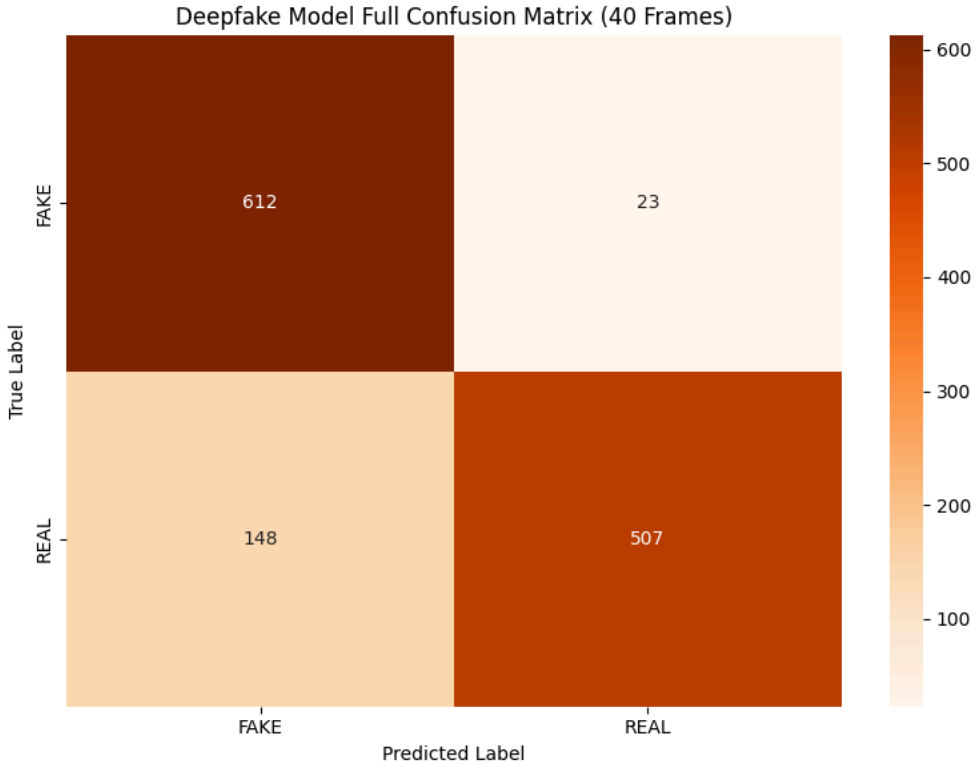

Figure 5.5.3 : Confusion Matrix (40 frames)

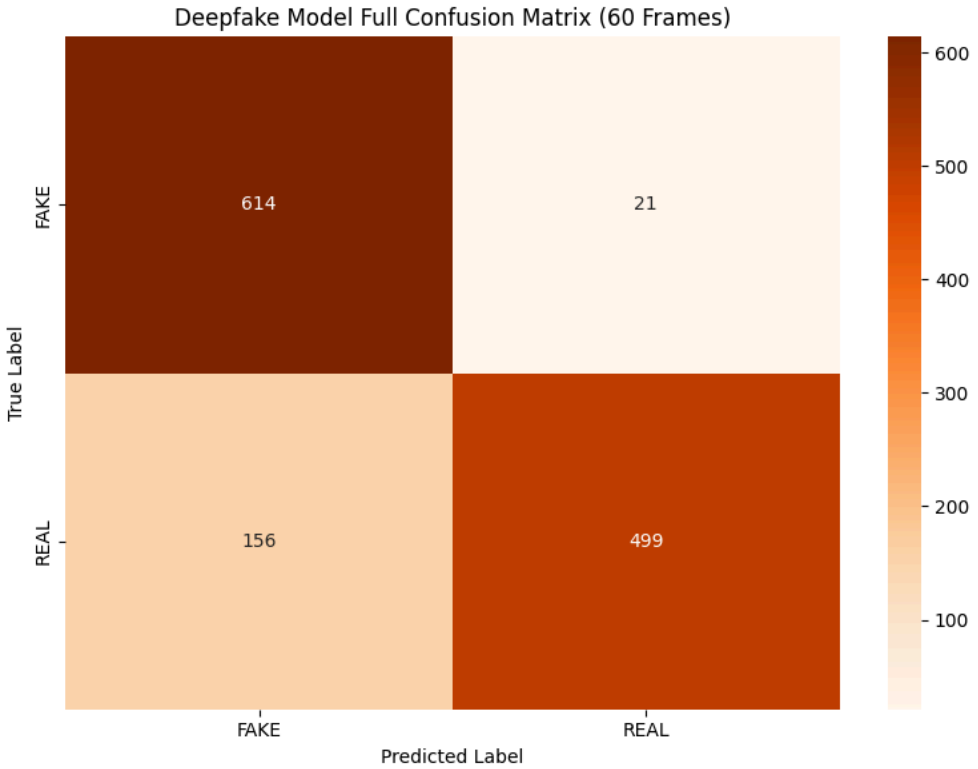

Figure 5.5.4 : Confusion Matrix (60 frames)

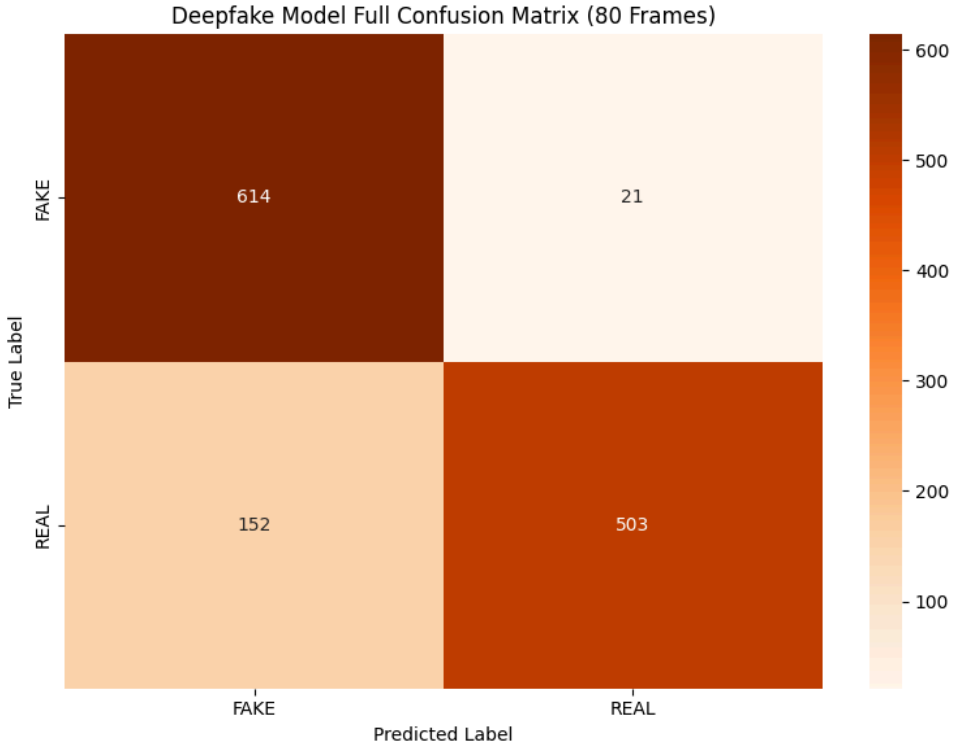

Figure 5.5.5 : Confusion Matrix (80 frames)

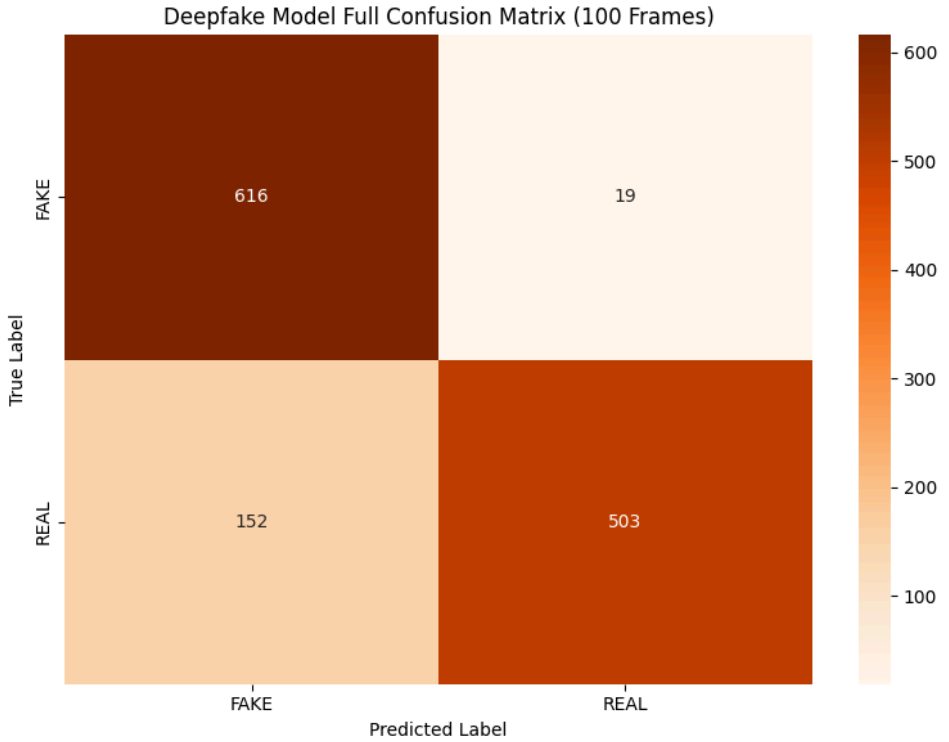

Figure 5.5.6 : Confusion Matrix (100 frames)

5.6 Screenshots

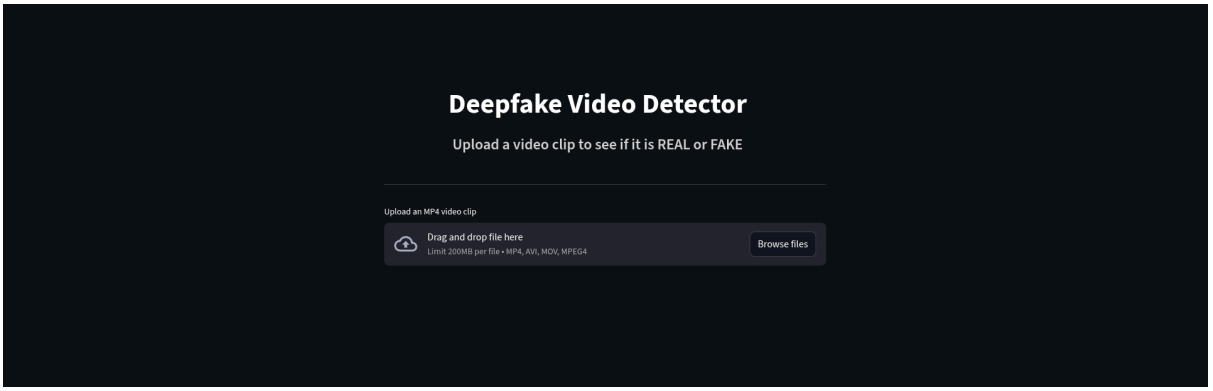

Figure 5.6.1 : Web Interface

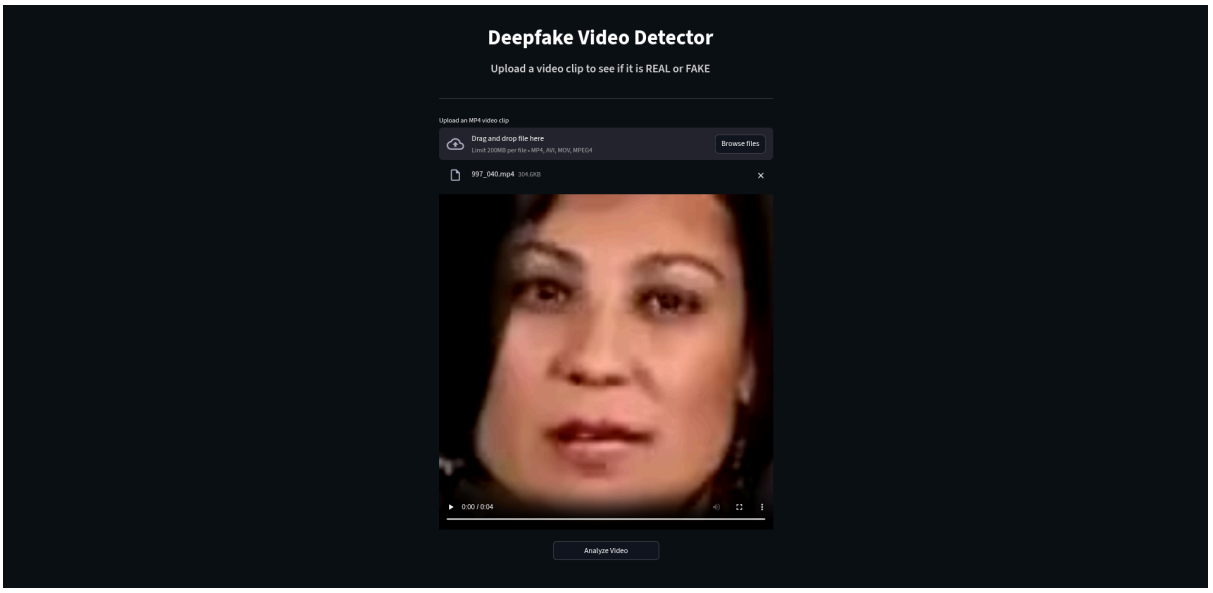

Figure 5.6.2 : Web Interface (Video uploaded)

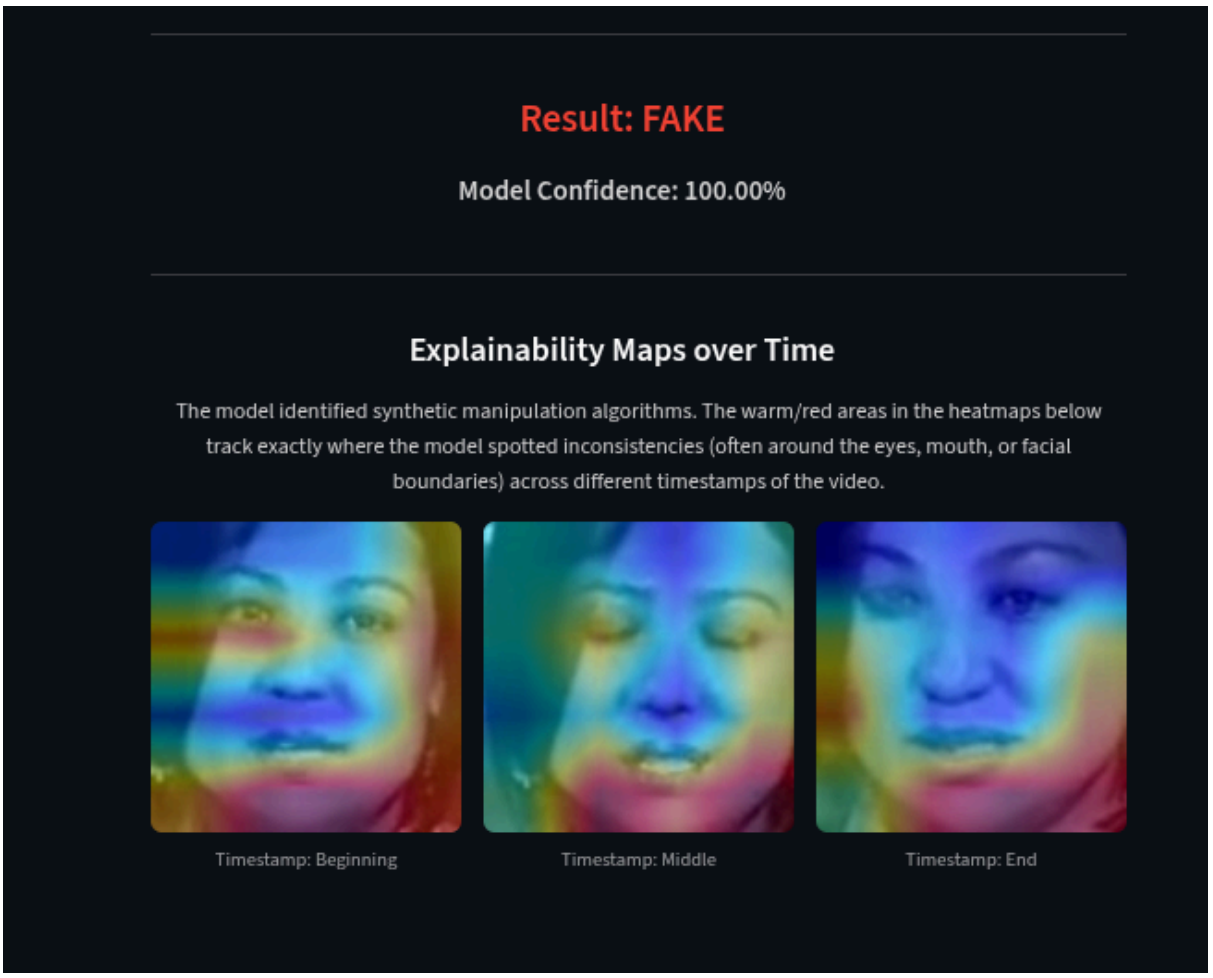

Figure 5.6.3 : Heatmap (Result : FAKE)

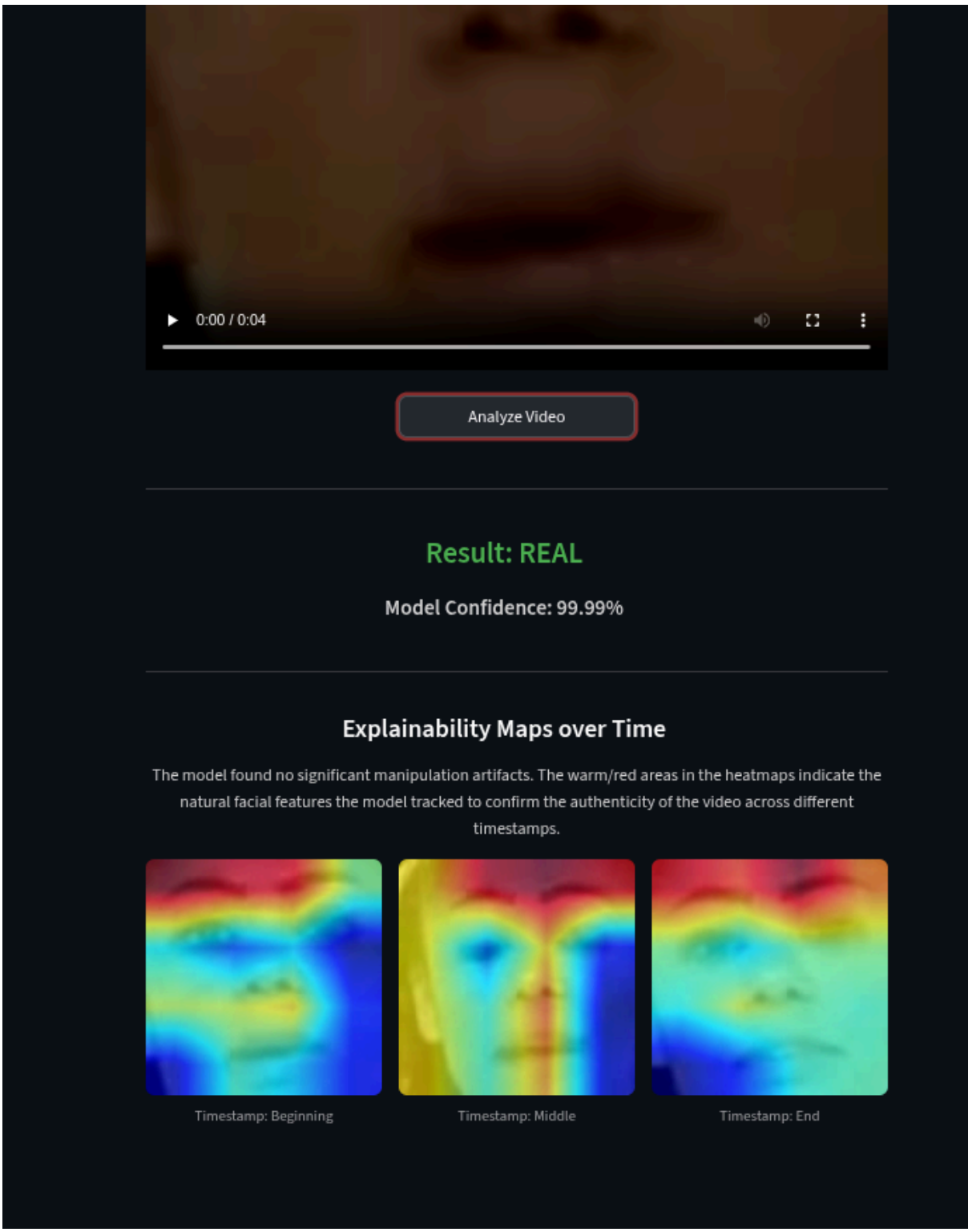

Figure 5.6.4 : Heatmap (Result : REAL)

5.7 Comparison with Existing Methods

Method	Accuracy	Limitation
CNN (XceptionNet)	~90–95%	No temporal analysis
MesoNet	~80–85%	Lower accuracy
CNN + RNN (existing)	~85–90%	Higher complexity
Proposed Model (CNN + LSTM)	86.74%	Moderate computation

Table 5.7.1 : Comparison

5.8 Analysis and Interpretation

The results of the experiment show that the suggested model is successful in identifying deepfake films, with a maximum accuracy of 86.74%. Up to 40 frames, it has been found that increasing the number of frames greatly enhances model performance; beyond that, the benefit becomes negligible.

Higher frame durations do not affect the precision and recall values, suggesting steady model performance with a balanced capacity to reduce false positives and false negatives. Strong overall classification performance is reflected in the F1-score, which also exhibits consistent improvement and reaches its maximum value at 100 frames.

One crucial finding is that employing more than 40 frames raises processing costs but does not considerably improve accuracy. Thus, 40 frames can be regarded as the best option for striking a compromise between efficiency and performance.

The suggested CNN + LSTM strategy improves detection performance over conventional frame-based models by efficiently combining spatial and temporal data. Although some models are more accurate, they frequently need more sophisticated designs and computer power, which makes the suggested method more useful for practical uses.

6. CONCLUSION AND FUTURE WORK

Conclusion

Through the analysis of both spatial and temporal data, a deep learning-based system was created in this study to identify deepfake films. The suggested method combines a Long Short-Term Memory (LSTM) network to capture temporal irregularities across video frames with a Convolutional Neural Network (ResNeXt-50) to extract frame-level characteristics.

A balanced dataset of real and false videos gathered from various sources was used to train and assess the system. The model's efficacy in detecting modified information is demonstrated by the experimental findings, which show an accuracy of 86.74%. When compared to conventional techniques that just examine individual frames, the use of combined spatial and temporal analysis improves performance.

The project emphasises how crucial deep learning is to solving contemporary cybersecurity issues, especially when it comes to identifying fake media. The created technique helps to increase security and confidence in digital platforms and offers a workable way to detect deepfakes.

Key Contributions

- Created a hybrid deep learning model by fusing LSTM and CNN.
- Developed a whole pipeline that includes prediction and video preparation.
- Reliability was attained on a balanced real-world dataset.
- Showed how well spatial and temporal information may be combined.
- Offered a scalable method for identifying deepfakes

Future Work

Although the suggested approach is effective in detecting deepfake movies, there are various areas for additional enhancement and extension. One potential approach is to improve the model architecture by combining sophisticated techniques like Vision Transformers (ViT) or experimenting with deeper and more optimised convolutional neural networks. Performance can also be increased by using better hyperparameter tuning and optimisation methodologies.

Furthermore, the dataset can be expanded to include a broader and more diversified group of movies, particularly real-world data with noise, compression artefacts, and varying quality levels. This would enhance the model's capacity to generalise to realistic settings.

Finally, the system can be transformed into a user-friendly application, such as a web or mobile platform, making it accessible to users. It may also be connected into social media platforms to automatically detect and block deepfake content, improving its real-world utility.

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